Sahaj Bhadja

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Summary

Software test engineer with 2+ years building **automation frameworks**, **CI/CD-integrated test suites**, and integration validation across **Python/Java/C++** codebases. Experienced in device/API testing, reproducible test artifacts, and quality practices for safety-conscious systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Software Engineering, Computer Networks, Database Systems, Operating Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Structures

Experience

USC Dornsife **Los Angeles, CA**
Software Test Engineer *September 2025 – Present*

- Designed **automated validation frameworks** for platform integrations serving 500+ users, including regression suites, data reconciliation checks, and release-readiness gates across dev/test/production environments.
- Produced structured **test logs and failure artifacts** for 5K+ monthly pipeline jobs, supporting root-cause analysis, deterministic replay, and CI/CD quality metrics at 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Test Engineering Intern *June 2025 – August 2025*

- Automated **web and API integration tests** for React/Flask systems handling 10K+ daily requests, validating end-to-end workflows, timing behavior, and defect reproduction with cross-functional engineering partners.
- Supported mobile-adjacent testing workflows using **ADB-style device commands** and scripted diagnostics to accelerate triage of customer-facing application issues.

USC AD Labs **Los Angeles, CA**
Software Engineering Assistant *January 2025 – May 2025*

- Built reusable **test harnesses and scenario libraries** across 100K+ records, enabling system/integration validation, regression comparisons, and documented verification evidence for research platforms.
- Collaborated with engineers on structured test reporting and quality reviews to improve traceability and repeatable verification practices.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Maintained **20+ automated PyTest suites** integrated into Jenkins CI/CD for distributed microservices, improving regression coverage and release confidence by **68%** across Windows/Linux environments.
- Developed **API and ETL integration tests** with mock services and fault-injection scenarios, reducing production defects and supporting hardware/software boundary debugging.

MedDevice Test Systems **India**
Test Automation Intern *January 2023 – April 2023*

- Implemented **Python automation libraries** for device communication testing, including **BLE connection lifecycle** scenarios (pairing, reconnect, drop/recover) and latency/staleness validation using simulators and mocks.
- Integrated automated suites into CI pipelines with standardized logs/reports suitable for regulated verification workflows and cross-team quality reviews.

Skills

Automation: Test framework design, reusable libraries, system/integration testing, mocks/simulators, regression suites
Languages & Tools: **Python**, **Java**, **C++**, ADB commands, Jenkins CI/CD, Git, PyTest, Mockito
Domains: Device/API protocols, BLE testing concepts, connected health workflows, quality-system awareness, artifact reporting

Projects

BLEInterop Test Suite | *Python, BLE Simulation, Test Automation*

- Built automation for device-to-device communication tests covering pairing, bonding, reconnect, and fault scenarios with deterministic logs for integration verification.

HealthSync Validator | *Java, Python, Integration Testing*

- Developed reusable test modules for data validation, timing checks, and backend/mobile API interoperability in connected-health style distributed systems.

APIValidator | *Python, REST APIs, CI/CD*

- Created API test harnesses with automated reporting and pipeline integration, supporting repeatable validation across firmware-adjacent service workflows.